

Professor, Korea University – Department of Electrical and Computer Engineering, Seoul, Korea

Visiting Professor, Seoul National University Hospital, Seoul, Korea

Director, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center, Korea University, Seoul, Korea

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Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
 - Ph.D. (08/2009–08/2014) in **Computer Science**, Mark and Mary Stevens School of Computing and Artificial Intelligence
Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group (Electrical Engineering))
 - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
 - M.S. (05/2012) in **Electrical Engineering**
 - **Korea University – College of Informatics** *Seoul, Republic of Korea*
 - M.S. (03/2004–02/2006) in **Computer Science and Engineering**
 - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
 - Professor (09/2019–Present)**, Department of Electrical and Computer Engineering
 - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
 - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
 - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
 - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
 - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
 - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
 - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
 - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
 - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
 - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
 - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
 - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
 - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
 - **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**
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University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
 - Principal Investigator, Software Star-Lab (07/2024–12/2031)**, Quantum AI Empowered Second-Life Platform Technology

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
 - Associate Dean (07/2026–Present)**, University College
 - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
 - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
 - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, Seoul, Republic of Korea
Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science
Member (03/2021–02/2027), Korea University 120th Anniversary Planning Committee
Member (06/2021–03/2022, 09/2022–08/2023), Academic Affair Planning Committee
Member (03/2021–08/2023), Distance Learning Management Committee
Steering Committee Member (07/2020–06/2022), Institute of Data Science (IDS)

Awards and Honors

Research and Academic Excellence (International)

- Top Cited Article in ETRI Journal (Wiley), among work published in 2024 2025
- Finalist, AAAI Student Abstract and Poster Session – Oral Presentation 2026 (Top 35), 2023 (Top 25)
- IEEE ICTC Best Paper Award – IEEE Communications Society 2022
- Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022) 2022
- IEEE MMTC Best Journal Paper Award – IEEE Communications Society 2021
- IEEE ICOIN Best Paper Award – IEEE Computer Society 2021
- IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society 2020
- IEEE Systems Journal Best Paper Award – IEEE Systems Council 2020
- Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation Q1/2015
- Annenberg Graduate Fellowship Award – University of Southern California 2009–2013
Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- IEEE Seoul Section Student Paper Contest Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, since 2016)

- HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS 2025
- Best Paper Award, The Journal of KICS – KICS 2024
- HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS 2023
- Haedong Young Scholar Award – KICS and Haedong Foundation 2018
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- Granite Tower Best Research Award (Top 3%), Totally, 2 times
– 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
– 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- Best Research Achievement Award – Korea University, School of Electrical Engineering 2024
- Insung Research Grant Award – Korea University 01/2023
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- Granite Tower Best Teaching Award (Top 5%), Totally, 7 times
– Probability and Random Process KECE209-02 (2025)
– Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
– Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
– Introduction to Computers SEMI103 (2021)
– Future Mobility Technology GEQR075 (2022)
- Best Teaching Award (Top 20%), Totally, 7 times
– Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
– Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
– Computer Language and Laboratory EGRN151-09 (2020)
– Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group 08/2025
- Certificate of Appreciation – IEEE/IFIP WiOpt (2024) 10/2024
- Best Editor Award – ICT Express (Elsevier) 07/2023
- 2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure 2022
- Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST) 2021
- Certificate of Appreciation – Department of Computer Science, University of Southern California 2010

Academic and University Services (Korea Regional)

- Outstanding Contribution Award
– The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
– Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
– Open Standards and ICT Association (OSIA) 2025, 2021
– The Institute of Electronics and Information Engineers (IEIE) 2025

- **Commemorative Plaque, KICS Excellent Lecture** (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing** (Group Leader) 2025
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021

Student Awards (International)

- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

R&D Projects

University-Wide/Center/Flagship Projects

- **(Center Director, PI) Net-Zero CAFE Research Center** (07/2024–12/2031) ITRC (Korea Univ)
- **(SW Star-Lab, PI) Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031) Star-Lab (Korea Univ)
- **Intelligent 6G Wireless Access System Research Center** (04/2021–12/2025) 6G AI Research Center (Korea Univ)
- **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (06/2021–05/2024) Basic Research Lab (Ajou)
- **Nano UAV Intelligence Systems Research Lab** (10/2020–08/2023) ADD Military Special Research Center (Kwangwoon)
- **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (06/2020–12/2022) ITRC (Hanyang)
- **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (08/2018–12/2021) ITRC (SNU-Hospital)
- **Novel Data Science Driven Framework for Efficient Network Design** (06/2017–05/2020) Basic Research Lab (Chung-Ang)
- **Intelligent Internet of Energy (IoE) Data Research Center** (02/2020–05/2020) ITRC (Kookmin)

Consortium Projects

- **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (09/2024–08/2027) IITP
- **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development** (04/2022–12/2026) IITP
- **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment** (04/2021–12/2025) IITP
- **Integrated Perception Technology Developments for Public Safety Platforms** (06/2019–05/2023) NRF
- **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm** (07/2019–12/2021) MHW
- **Virtual Presence in Moving Objects through 5G (PriMO-5G)** (06/2018–06/2021) IITP
- **Distributed Secure Platform for Scalable Clinical OMOP CDM Models** (04/2019–12/2020) MHW

Individual Projects

- **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System** (03/2025–02/2028) NRF
- **Research on Trainability and Learnability in Quantum Machine Learning Models** (03/2026–10/2027) ETRI-Affiliated Institute
- **Advancement Technology Development for Torpedo Deception Strategies in Submarines** (11/2022–11/2026) LIG Nex1
- **Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence** (11/2022–11/2026) LIG Nex1
- **Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation** (05/2026–10/2026) KOITA
- **mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners** (06/2026–09/2026) LG Electronics (Robotics Lab)
- **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers** (09/2020–11/2025) Samsung Electronics
- **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning** (05/2025–11/2025) ETRI
- **Discovering Quantum-Advantage Cases for Industry Adoption** (09/2025–11/2025) KTL
- **Research on Generative Quantum Machine Learning Models** (04/2025–10/2025) ETRI-Affiliated Institute
- **Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility** (03/2022–02/2025) NRF
- **Research on Learning-based Swarm Mission Planning Algorithms** (03/2024–02/2025) LIG Nex1
- **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks** (05/2024–11/2024) ETRI
- **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies** (10/2023–09/2024) NRF
- **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration** (12/2022–04/2024) Hyundai Motors
- **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks** (09/2023–11/2023) ETRI
- **Routing Algorithms for LEO Satellite Networks** (12/2022–08/2023) Solvit System
- **Optimal Positioning Algorithms for Wide-Area Relaying Networks** (12/2022–08/2023) Solvit System
- **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data** (11/2022–02/2023) Cipherome
- **Cellular/Wi-Fi Handover Technology Development** (02/2022–12/2022) LG Electronics CTO Division (Smart Mobility Lab)
- **Autonomous Intelligent COA Search Methods for Cyber-Attacks** (12/2021–11/2022) ADD
- **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA** (10/2019–04/2022) NRF
- **Research Trends in Digital Twin Applications to Autonomous Driving** (03/2022–04/2022) Hyundai NGV
- **Distributed Learning System Design and Implementation for Clinical Applications** (02/2022–03/2022) Cipherome
- **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving** (06/2019–02/2022) NRF
- **Research on Intelligent Agent-based CPS Security and Reliability** (04/2021–11/2021) TTA
- **Multi-GPU based Automotive HPC Platform Development** (04/2020–10/2020) ETRI
- **Super-Resolution Performance Optimization in Mobile Platforms** (05/2020–08/2020) Samsung SDS
- **Cooperative Deep Reinforcement Learning for Online Game Multi-Agents** (04/2020–08/2020) ETRI
- **Deep Learning Algorithms for mVOC Concentration Analysis** (03/2020–06/2020) Samsung Electronics (C-Lab)
- **Visual Recognition Software Implementation using Deep Learning Tools** (05/2019–11/2019) Hyundai Motors
- **Verification Testbed Implementation for Privacy-Preserving Trust Data Generation** (10/2019–11/2019) ETRI

- *Measurement and Analysis of Multi-Task GPU Scheduling Delays* (05/2019–10/2019) **ETRI**
- *mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services* (06/2016–05/2019) **NRF**
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) **KEPCO Research Institute**
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) **ETRI**
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) **ETRI**
- *A Priori Techniques Research for Efficient Multi-Edge Computing* (06/2017–12/2017) **Samsung Electronics (Software Center)**
- *Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms* (04/2017–12/2017) **IITP**
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) **ETRI**

University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* **Intel Labs, Verizon Wireless, Cisco Systems**
- *60 GHz Real-Time Wireless Video Broadcasting* **Disney Research Zürich**
- *Annenberg Graduate Fellowship Award* **University of Southern California**

Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) **PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)**
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) **KU University College**
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) **KU Insung Research Grant**
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) **KU Future Research Grant**

Selected Publications

- **12,204+ Citations** (H-index: 54+, i10-index: 257+), obtained from Google Scholar Profile (as of July 26, 2026)
- **Journals and Magazines: 174** publications; among them, **130** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, [J. Kim](#))

Selected Papers

■ Top-Tier Conference Publications

- [ICML'26]** *Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity*
ICML 2026 (H. K. Lee, [J. Kim](#), S. W. Yoon) (Acceptance Rate: 26.56%)
- [ACL'26]** *Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom*
ACL 2026 (J. Lee, G. Youn, S. Lee, C. Im, [J. Kim](#), C. Yoo)
- [CIKM'25]** *Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models*
ACM CIKM 2025 (E. J. Roh, [J. Kim](#)) (Acceptance Rate: 30.63%)
- [CIKM'25]** *Filtered One-Shot Training for Quantum Architecture Search*
ACM CIKM 2025 (S. B. Son, S. Y.-C. Chen, [J. Kim](#), S. Park) (Acceptance Rate: 30.63%)
- [CIKM'25]** *LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data*
ACM CIKM 2025 (G. Youn, J. Lee, [J. Kim](#), C. Yoo) (Acceptance Rate: 30.63%)
- [IPDPS'25]** *AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming*
IEEE IPDPS 2025 (X. Piao, J. Shim, [J. Kim](#), J. Kim) (Acceptance Rate: 24.71%)
- [WiOpt'25]** *Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach*
IEEE/IFIP WiOpt 2025 (G. S. Kim, J. Chung, T. Q. Duong, S. Park, [J. Kim](#))
- [NOMS'25]** *Joint Multi-Agent Reinforcement Learning and Message-Passing for Distributed Multi-UAV Network Management using Conflict Graphs*
IEEE/IFIP NOMS 2025 (Y. Cho, H. Lee, S. Park, [J. Kim](#))
- [CIKM'24]** *Hands-On Introduction to Quantum Machine Learning*
ACM CIKM 2024 (Tutorial) (S. Y.-C. Chen, [J. Kim](#))
- [WiOpt'24]** *Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management*
IEEE/IFIP WiOpt 2024 (S. Lee, G. S. Kim, S. Park, [J. Kim](#))
- [CIKM'23]** *Quantum Split Learning for Privacy-Preserving Information Management*
ACM CIKM 2023 (S. Park, H. Baek, [J. Kim](#)) (Acceptance Rate: 27.44%)
- [CIKM'23]** *Logarithmic Dimension Reduction for Quantum Neural Networks*
ACM CIKM 2023 (H. Baek, S. Park, [J. Kim](#)) (Acceptance Rate: 27.44%)
- [AAAI'23]** *Quantum Multi-Agent Meta Reinforcement Learning*
AAAI 2023 (W. Yun, J. Park, [J. Kim](#)) (Acceptance Rate: 19.61%)
- [CIKM'22]** *Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly*
ACM CIKM 2022 (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, [J. Kim](#)) (Acceptance Rate: 29.64%)
- [WiOpt'22]** *Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution*
IEEE/IFIP WiOpt 2022 (M. Choi, W. Yun, [J. Kim](#))
- [INFOCOM'22]** *Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks*
IEEE INFOCOM 2022 (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, [J. Kim](#)) (Acceptance Rate: 19.93%)

- [ICDCS'20] *Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications*
IEEE ICDCS 2020 (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen) (Acceptance Rate: 17.98%)
- [IJCAI'19] *Randomized Adversarial Imitation Learning for Autonomous Driving*
IJCAI 2019 (M. Shin, J. Kim) (Acceptance Rate: 17.89%)
- [ICDCS'18] *ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture*
IEEE ICDCS 2018 (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang) (Acceptance Rate: 20.63%)
- [ACM-MM'17] *REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints*
ACM Multimedia 2017 (J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi) (Acceptance Rate: 27.63%)
- [MobiSys'10] *Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones*
ACM MobiSys 2010 (J. Paek, J. Kim, R. Govindan) (Acceptance Rate: 19.84%)

■ Magazines and Journals

◀ Minor ▶

- [IOT].minor] *USD2C: UAV-Assisted Secure Data Collection and Computation Framework for Internet of Multimedia Things (IoMT) Applications*, **IEEE Internet of Things Journal**, v(n):ppp-ppp (I. Ahmad, J. Kim, S. Shin)
- [TSUSC].minor] *Dynamic Sun Tracking Control for Efficient Solar Power Charging in Sustainable Cube-Satellites*, **IEEE Transactions on Sustainable Computing**, v(n):ppp-ppp (G. S. Kim, S. Park, J. Kim)
- [AC].minor] *Physics-Validated Action Masking with LLM-Guided Decomposition for Autonomous Excavation*, **Automation in Construction**, v(n):ppp-ppp (J. Cho, M. Shin, B. Kim, B. Cha, J. Kim, S. Jung)

◀ Accepted ▶

- [CIM.accepted] *Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces*, **IEEE Computational Intelligence Magazine**, v(n):ppp-ppp (S. B. Son, J. Kim, S. Park)
- [CM.accepted] *Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control*, **IEEE Communications Magazine**, v(n):ppp-ppp (H. Lee, S. Park, S. Jung, J. Kim)
- [CM.accepted] *Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective*, **IEEE Communications Magazine**, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)
- [NM.accepted] *Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks*, **IEEE Network**, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)
- [INM.accepted] *Toward Quantum Combinatorial Optimization: From QAOA Fundamentals to Industrial Deployment*, **IEEE Nanotechnology Magazine**, v(n):ppp-ppp (Y. P. Hong, J. Gil, D. Kim, S. Oh, B. S. Kim, J. Kim)
- [TMC.accepted] *Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments*, **IEEE Transactions on Mobile Computing**, v(n):ppp-ppp (G. S. Kim, E. J. Roh, S. Jung, S. Park, J. Kim)
- [TMC.accepted] *Efficient Quantum Neural Architecture Search with Filtered Supernet for Mobile Learning Systems*, **IEEE Transactions on Mobile Computing**, v(n):ppp-ppp (S. B. Son, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)

◀ 2026 ▶

- [TVT'26.10] *Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations*, **IEEE Transactions on Vehicular Technology**, 75(10):ppp-ppp (S. Kim, J. Kim, J. Kim, S. Jung)
- [FGCS'26.09] *GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems*, **Future Generation Computer Systems**, 182:108485 (X. Piao, J. Kim, J.-K. Kim)
- [IOT]'26.08] *Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks*, **IEEE Internet of Things Journal**, 13(16):ppp-ppp (G. S. Kim, C. Im, Y. G. Kim, J. Ha, J. Kim)
- [EAAI'26.08] *A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management: Fundamentals, Challenges and Opportunities*, **Engineering Applications of Artificial Intelligence**, 177:114991 (M. Jha, S. Chen, C. Kulkarni, J. Kim)
- [TMC'26.07] *Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks*, **IEEE Transactions on Mobile Computing**, 25(7):10385–10401 (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)
- [Access'26.07] *Two-Stage MPC-Guided Imitation and Reinforcement Learning for Autonomous Underwater Vehicle Control in Dynamic Obstacle Environments*, **IEEE Access**, 14:99636–99648 (S. Kim, E. J. Roh, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)
- [IOT]'26.06] *Quantum Federated Gradient Aggregation Using the Parameter-Shift Rule*, **IEEE Internet of Things Journal**, 13(11):25367–25379 (J. Chung, C. Im, S. Park, J. Kim, W. Lee)
- [Nature'26.05] *Fourier Analysis Perspective on Quantum Neural Networks*, **Communications Physics (Nature)**, 9:176 (S. Oh, E. J. Roh, A. Vasilakos, S. Park, J. Kim)
- [TVT'26.04] *Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control*, **IEEE Transactions on Vehicular Technology**, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, J. Kim)
- [TAES'26.03] *Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket*, **IEEE Transactions on Aerospace and Electronic Systems**, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)
- [IOT]'26.03] *Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks*, **IEEE Internet of Things Journal**, 13(6):12322–12339 (J. Jang, I.-S. Cho, M. Shin, J. Kim, S. Jung)
- [IOT]'26.02] *Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch*, **IEEE Internet of Things Journal**, 13(3):5379–5392 (H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, J. Kim)
- [PRA'26.02] *Understanding Qubit Trajectories in Parametrized Quantum Circuits*, **Physical Review A**, 113:022609 (S. Oh, C. Im, S. Park, J. Kim)
- [AC'26.02] *Collaborative Learning Architecture for Autonomous Excavator Planning and Execution*, **Automation in Construction**, 182:106742 (J. Cho, M. Shin, J. Kim, S. Jung)
- [TAES'26.01] *Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles*, **IEEE**

Transactions on Aerospace and Electronic Systems, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, J. Kim)

[TMC'26.01] Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks, *IEEE Transactions on Mobile Computing*, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, J. Kim)

◀ 2025 ▶

[MM'25.10-12] Quantum Jump to Virtual Worlds: High-Quality Multiple Virtual Meta-Space Realization in Metaverse, *IEEE MultiMedia*, 32(4):37–44 (S. Park, J. Kim)

[TIV'25.11] Adaptive Quantum Federated Learning for Autonomous Surveillance Multi-Drone Networks, *IEEE Transactions on Intelligent Vehicles*, 10(11):5055–5059 (S. Park, C. Park, S. Jung, J. Kim)

[JS'25.10] Joint Scalable Quantum Convolutional Neural Network and Reverse-Fidelity Training for High-Accurate Recognition in Unmanned Aerial Vehicle Surveillance, *The Journal of Supercomputing*, 81(16):1495 (E. J. Roh, J. Kim, S. Jung, S. Park)

[TASE'25.09] Adaptive Excavation Automation in Complex Soil Environments using Reinforcement Learning, *IEEE Transactions on Automation Science and Engineering*, 22:21181–21194 (M. Shin, J. Cho, J. Kim, S. Jung)

[ESWA'25.09] Correlation-Assisted Spatio-Temporal Reinforcement Learning for Stock Revenue Maximization, *Expert Systems with Applications*, 289:128361 (J. Chung, M. Kim, S. Min, H. Choi, S. Park, J. Kim)

[CIM'25.08] Quantum-Eyes: Scalable Quantum Convolutional Neural Networks for Low-Overhead Object Detection, *IEEE Computational Intelligence Magazine*, 20(3):63–74 (J. Kim, E. J. Roh, C. Im, S. Park)

[TON'25.08] Slimmable Federated Reinforcement Learning for Energy-Efficient Proactive Caching, *IEEE Transactions on Networking*, 33(4):2079–2094 (H. Baek, G. S. Kim, S. Park, A. F. Molisch, J. Kim)

[TAES'25.08] Quantum Multiagent Reinforcement Learning for Joint Cube-Satellites and High-Altitude Long-Endurance Aerial Vehicles in SAGIN, *IEEE Transactions on Aerospace and Electronic Systems*, 61(4):9490–9510 (G. Kim, Y. Cho, S. Park, S. Jung, J. Kim)

[IOTJ'25.08] Joint Interference Approximation and Guard-Band Management for Spectrum-Efficient Integrated NTN-TN Networks, *IEEE Internet of Things Journal*, 12(15):32220–32236 (J. Jang, J. Kim, J. Kim, S. Jung)

[IOTJ'25.08] Carbon-Aware Edge Computing for Internet of Everything Networks: A Digital Twin Approach, *IEEE Internet of Things Journal*, 12(15):29240–29251 (D. V. Huynh, S. R. Khosravirad, V. Sharma, J. Kim, B. Canberk, T. Q. Duong)

[JCN'25.08] Stabilized Classification Control using Multi-Stage Quantum Convolutional Neural Networks for Autonomous Driving, *IEEE/KICS Journal of Communications and Networks*, 27(4):264–272 (E. J. Roh, S. Park, S. Jung, J. Kim)

[JCN'25.08] Special Issue on Quantum Technologies for Communication Systems, *IEEE/KICS Journal of Communications and Networks*, 27(4):217–221 (S. Y. Shin, Z. Han, S. Ali, S. Y.-C. Chen, Y. Liu, S. Park, J. Kim)

[IOTJ'25.07] Quantum Reinforcement Learning for Lightweight LEO Satellite Routing, *IEEE Internet of Things Journal*, 12(14):28986–29004 (G. S. Kim, S. Lee, I.-S. Cho, S. Park, J. Kim)

[IOTJ'25.07] Privacy-Preserving Uncertainty Calibration using Perceptual Encryption in Cloud-Edge Collaborative Artificial Intelligence of Things, *IEEE Internet of Things Journal*, 12(13):25424–25441 (I. Ahmad, J. Kim, S. Shin)

[IOTJ'25.07] Joint Quantum Reinforcement Learning and Neural Myerson Auction for High-Quality Digital-Twin Services in Multitier Networks, *IEEE Internet of Things Journal*, 12(13):23722–23735 (S. Park, G. S. Kim, J. Kim)

[NN'25.07] Quantum Federated Learning with Pole-Angle Quantum Local Training and Trainable Measurement, *Neural Networks*, 187:107301 (S. Park, H. Lee, S. B. Son, S. Jung, J. Kim)

[JS'25.07] Quantum Infidelity Codistillation for Fast and Accurate Distributed Quantum Machine Learning, *The Journal of Supercomputing*, 81(10):1151 (S. Oh, J. Kim, J. Park, H. Baek, H. Lee, J. Kim, S.-L. Kim)

[IOTJ'25.06] Entanglement-Controlled Quantum Federated Learning, *IEEE Internet of Things Journal*, 12(11):18318–18330 (S. Park, H. Lee, S. Jung, J. Park, M. Bennis, J. Kim)

[JS'25.06] SQUAD: Software Testing for Quantum Distributed Learning Software, *The Journal of Supercomputing*, 81(9):1071 (S. Park, J. H. Cho, H. J. Yook, G. S. Jhun, Y. K. Lee, J. Kim, S. Park)

[TVT'25.05] Dynamic Quantum Federated Learning for UAV-based Autonomous Surveillance, *IEEE Transactions on Vehicular Technology*, 74(5):8158–8170 (S. Park, S. B. Son, S. Jung, J. Kim)

[TNSM'25.04] Intelligent Extra Resource Allocation for Cooperative Awareness Message Broadcasting in Cellular-V2X Networks, *IEEE Transactions on Network and Service Management*, 22(2):1677–1689 (S. Jung, J.-H. Kim, J. Kim)

[TMC'25.02] Fast Quantum Convolutional Neural Networks for Low-Complexity Object Detection in Autonomous Driving Applications, *IEEE Transactions on Mobile Computing*, 24(2):1031–1042 (E. J. Roh, H. Baek, D. Kim, J. Kim)

[JS'25.02] Hybrid Quantum-Classical 3D Object Detection using Multi-Channel Quantum Convolutional Neural Network, *The Journal of Supercomputing*, 81(3):455 (E. J. Roh, J. Y. Shim, J. Kim, S. Park)

[EL'25.01] Fast Batch Gradient Descent in Quantum Neural Networks, *IET Electronics Letters*, 61(1):e70162 (J. Y. Shim, J. Kim)

◀ 2024 ▶

[CM'24.12] The Matrix: Quantum AI for Interacting Two Worlds in Prioritized Metaverse Spaces, *IEEE Communications Magazine*, 62(12):97–103 (S. Park, H. Baek, J. Kim)

[TON'24.12] Spatio-Temporal Multi-Metaverse Dynamic Streaming for Hybrid Quantum-Classical Systems, *IEEE/ACM Transactions on Networking*, 32(6):5279–5294 (S. Park, H. Baek, J. Kim)

[TMC'24.12] Joint Quantum Reinforcement Learning and Stabilized Control for Spatio-Temporal Coordination in Metaverse, *IEEE Transactions on Mobile Computing*, 23(12):12410–12427 (S. Park, J. Chung, C. Park, S. Jung, M. Choi, S. Cho, J. Kim)

[IOTJ'24.12] Markov Decision Policies for Distributed Angular Routing in LEO Mobile Satellite Constellation Networks, *IEEE Internet of Things Journal*, 11(23):38744–38754 (S. Park, G. S. Kim, S. Jung, J. Kim)

[TIV'24.11] Neural Myerson Auction for Truthful and Distributed Mobile Charging in UAV-Assisted Digital-Twin Networks, *IEEE Transactions on Intelligent Vehicles*, 9(11):7326–7338 (S. Jung, H. Baek, J. Kim)

[CM'24.10] Quantum Multi-Agent Reinforcement Learning is All You Need: Coordinated Global Access in Integrated TN/NTN Cube-Satellite Networks, *IEEE Communications Magazine*, 62(10):86–92 (S. Park, G. S. Kim, Z. Han, J. Kim)

- [Access'24.10] Sensing-to-Sky Intermittent Connectivity Realization for LTE-Enabled Drone Platforms: Embedded Design, Measurement Study, and Positioning Applications, *IEEE Access*, 12:137360–137372 (J. Kim, S. Park, U. Jo, T. Kim, S. Jung, J. Kim)
- [FGCS'24.10] AQUA: Analytics-driven Quantum Neural Network (QNN) User Assistance for Software Validation, *Future Generation Computer Systems*, 159:545–556 (S. Park, H. Baek, J. W. Yoon, Y. K. Lee, J. Kim)
- [ETRI'24.10] Trends in Quantum Reinforcement Learning: State-of-the-Arts and the Road Ahead, *ETRI Journal*, 46(5):748–758 (S. Park, J. Kim) **(Top Cited Article, among work published in 2024)**
- [TNSM'24.08] Cooperative Multi-UAV Positioning for Aerial Internet Service Management: A Multi-Agent Deep Reinforcement Learning Approach, *IEEE Transactions on Network and Service Management*, 21(4):3797–3812 (J. Kim, S. Park, S. Jung, C. Cordeiro)
- [Access'24.08] Enhancing Cost-Effective 5G Virtualized RAN Pooling Gain on Clouds: An Intelligent Auto-Scaling Approach, *IEEE Access*, 12:111322–111333 (K. Cho, J. Kim, S. Jung)
- [TVT'24.07] Age-of-Information Aware Caching and Delivery for Infrastructure-Assisted Connected Vehicles, *IEEE Transactions on Vehicular Technology*, 73(7):10681–10696 (S. Park, C. Park, S. Jung, M. Choi, J. Kim)
- [CM'24.06] Quantum Multi-Agent Reinforcement Learning for Autonomous Mobility Cooperation, *IEEE Communications Magazine*, 62(6):106–112 (S. Park, J. P. Kim, C. Park, S. Jung, J. Kim)
- [TVT'24.04] Learning-Based Cooperative Mobility Control for Autonomous Drone-Delivery, *IEEE Transactions on Vehicular Technology*, 73(4):4870–4885 (S. Park, C. Park, J. Kim)
- [Access'24.04] Dynamic Quantum Federated Learning for Satellite-Ground Integrated Systems using Slimmable Quantum Neural Networks, *IEEE Access*, 12:58239–58247 (S. Park, S. Jung, J. Kim)
- [Access'24.04] Quantum Reinforcement Learning for Spatio-Temporal Prioritization in Metaverse, *IEEE Access*, 12:54732–54744 (S. Park, H. Baek, J. Kim)
- [TWC'24.03] Joint User Clustering, Beamforming, and Power Allocation for mmWave-NOMA with Imperfect SIC, *IEEE Transactions on Wireless Communications*, 23(3):2025–2038 (B. Lim, W. Yun, J. Kim, Y.-C. Ko)
- [TGCN'24.03] Joint Delay-Sensitive and Power-Efficient Quality Control of Dynamic Video Streaming using Adaptive Super-Resolution, *IEEE Transactions on Green Communications and Networking*, 8(1):103–117 (M. Choi, W. Yun, S. Son, S. Park, J. Kim)
- [TIV'24.02] Intelligent Caching for Seamless High-Quality Streaming in Vehicular Networks: A Multi-Agent Reinforcement Learning Approach, *IEEE Transactions on Intelligent Vehicles*, 9(2):3672–3686 (M. Choi, T. Xiang, J. Kim)
- [TNNLS'24.02] Hierarchical Deep Reinforcement Learning-based Propofol Infusion Assistant Framework in Anesthesia, *IEEE Transactions on Neural Networks and Learning Systems*, 35(2):2510–2521 (W. Yun, M. Shin, D. Mohaisen, K. Lee, J. Kim)
- [TMC'24.01] Learning Location from Shared Elevation Profiles in Fitness Apps: A Privacy Perspective, *IEEE Transactions on Mobile Computing*, 23(1):581–596 (U. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen)
- ◀ 2023 ▶
- [TON'23.12] SlimFL: Federated Learning with Superposition Coding over Slimmable Neural Networks, *IEEE/ACM Transactions on Networking*, 31(6):2499–2514 (W. Yun, Y. Kwak, H. Baek, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim)
- [EL'23.12] Two-Stage Architectural Fine-Tuning for Neural Architecture Search in Efficient Transfer Learning, *IET Electronics Letters*, 59(24):e13066 (S. Park, S. B. Son, Y. K. Lee, S. Jung, J. Kim)
- [IOTJ'23.11] Quantum Multiagent Actor–Critic Networks for Cooperative Mobile Access in Multi-UAV Systems, *IEEE Internet of Things Journal*, 10(22):20033–20048 (C. Park, W. Yun, J. P. Kim, S. Park, T. K. Rodrigues, S. Jung, J. Kim)
- [TVT'23.11] Two-Stage Self-Adaptive Task Outsourcing Decision Making for Edge-Assisted Multi-UAV Networks, *IEEE Transactions on Vehicular Technology*, 72(11):14889–14905 (S. Jung, C. Park, M. Levorato, J.-H. Kim, J. Kim)
- [ETRI'23.10] Joint Frame Rate Adaptation and Object Recognition Model Selection for Stabilized Unmanned Aerial Vehicle Surveillance, *ETRI Journal*, 45(5):811–821 (G. S. Kim, H. Lee, S. Park, J. Kim)
- [ETRI'23.10] Two Tales of Platoon Intelligence for Autonomous Mobility Control: Enabling Deep Learning Recipes, *ETRI Journal*, 45(5):735–745 (S. Park, H. Lee, C. Park, S. Jung, M. Choi, J. Kim)
- [ETRI'23.10] Special Issue on Autonomous Unmanned Aerial/Ground Vehicles and their Applications, *ETRI Journal*, 45(5):731–734 (J. Kim, Y.-C. Lee, J. H. Lee, J. S. Choi)
- [IC'23.09-10] EQuaTE: Efficient Quantum Train Engine for Run-Time Dynamic Analysis and Visual Feedback in Autonomous Driving, *IEEE Internet Computing*, 27(5):24–31 (S. Park, H. Feng, C. Park, Y. K. Lee, S. Jung, J. Kim)
- [OJCS'23.09] Real-Time High-Quality Visualization for Volumetric Contents Rendering: A Lyapunov Optimization Framework, *IEEE Open Journal of the Computer Society*, 4:243–252 (H. Baek, R. Lee, S. Jung, J. Kim, S. Park)
- [TIV'23.08] Multi-Agent Reinforcement Learning for Cooperative Air Transportation Services in City-Wide Autonomous Urban Air Mobility, *IEEE Transactions on Intelligent Vehicles*, 8(8):4016–4030 (C. Park, G. S. Kim, S. Park, S. Jung, J. Kim)
- [NN'23.08] Stereoscopic Scalable Quantum Convolutional Neural Networks, *Neural Networks*, 165:860–867 (H. Baek, W. Yun, S. Park, J. Kim)
- [IOTJ'23.06] Quantum Multiagent Actor-Critic Neural Networks for Internet-Connected Multirobot Coordination in Smart Factory Management, *IEEE Internet of Things Journal*, 10(11):9942–9952 (W. Yun, J. P. Kim, S. Jung, J.-H. Kim, J. Kim)
- [ICTE'23.06] Quantum Distributed Deep Learning Architectures: Models, Discussions, and Applications, *ICT Express*, 9(3):486–491 (Y. Kwak, W. Yun, J. P. Kim, H. Cho, J. Park, M. Choi, S. Jung, J. Kim)
- [Access'23.05] Entropy-Aware Similarity for Balanced Clustering: A Case Study with Melanoma Detection, *IEEE Access*, 11:46892–46902 (S. B. Son, S. Park, J. Kim)
- [ComNet'23.04] Self-Adaptive End-to-End Resource Management for Real-Time Monitoring in Cyber-Physical Systems, *Computer Networks*, 225:109669 (H.-C. Jo, H.-W. Jin, J. Kim)
- [ComNet'23.04] Truthful and Performance-Optimal Computation Outsourcing for Aerial Surveillance Platforms via Learning-based Auction, *Computer Networks*, 225:109651 (S. Jung, J.-H. Kim, D. Mohaisen, J. Kim)
- [CIBM'23.04] Deep Reinforcement Learning-based Propofol Infusion with a 3,000-subject Dataset in Anesthesia, *Computers in Biology and Medicine*, 156:106739 (W. Yun, M. Shin, S. Jung, J. Ko, H.-C. Lee, J. Kim)

- [Access'23.03] Audio-to-Visual Cross-Modal Generation of Birds, *IEEE Access*, 11:27719–27729 (J. Y. Shim, J. Kim, J.-K. Kim)
- [Access'23.02] Workload-Aware Scheduling using Markov Decision Process for Infrastructure-Assisted Learning-Based Multi-UAV Surveillance Networks, *IEEE Access*, 11:16533–16548 (S. Park, C. Park, S. Jung, J.-H. Kim, J. Kim)
- [TITS'23.01] Self-Configurable Stabilized Real-Time Detection Learning for Autonomous Driving Applications, *IEEE Transactions on Intelligent Transportation Systems*, 24(1):885–890 (W. Yun, S. Park, J. Kim, D. Mohaisen)
- ◀ 2022 ▶
- [JCN'22.12] Neural Myerson Auction for Truthful and Energy-Efficient Autonomous Aerial Data Delivery, *IEEE/KICS Journal of Communications and Networks*, 24(6):730–741 (H. Lee, S. Kwon, S. Jung, J. Kim)
- [JCN'22.12] Parallelized and Randomized Adversarial Imitation Learning for Safety-Critical Self-Driving Vehicles, *IEEE/KICS Journal of Communications and Networks*, 24(6):710–721 (W. Yun, M. Shin, S. Jung, S. Kwon, J. Kim)
- [TII'22.10] Cooperative Multi-Agent Deep Reinforcement Learning for Reliable Surveillance via Autonomous Multi-UAV Control, *IEEE Transactions on Industrial Informatics*, 18(10):7086–7096 (W. Yun, S. Park, J. Kim, M. Shin, S. Jung, D. Mohaisen, J.-H. Kim)
- [ICTE'22.09] Trustworthy Handover in LEO Satellite Mobile Networks, *ICT Express*, 8(3):432–437 (S. Jung, M. Lee, J. Kim, M. Yun, J. Kim, J. Kim)
- [TVT'22.07] Joint Pilot Design and Channel Estimation using Deep Residual Learning for Multi-Cell Massive MIMO under Hardware Impairments, *IEEE Transactions on Vehicular Technology*, 71(7):7599–7612 (B. Lim, W. Yun, J. Kim, Y.-C. Ko)
- [ITU'22.07] Dynamic Resource Scheduling for Real-Time Group Broadcasting in 6G Cellular Vehicular Networks, *ITU Journal on Future and Evolving Technologies*, 3(1):81–88 (S. Jung, M. Levorato, J. Kim)
- [ISJ'22.06] Securing Heterogeneous IoT with Intelligent DDoS Attack Behavior Learning, *IEEE Systems Journal*, 16(2):1974–1983 (N.-N. Dao, T. Phan, U. Sa'ad, J. Kim, T. Bauschert, D.-T. Do, S. Cho)
- [CSM'22.06] Recent and Future Evolution of Wi-Fi, *IEEE Communications Standards Magazine*, 6(2):8–11 (E. Au, L. Wilhelmsson, T. Baykas, J. Kim)
- [TMC'22.05] Supremo: Cloud-Assisted Low-Latency Super-Resolution in Mobile Devices, *IEEE Transactions on Mobile Computing*, 21(5):1847–1860 (J. Yi, S. Kim, J. Kim, S. Choi)
- [TVT'22.05] Stabilized Detection Accuracy Maximization using Adaptive SAR Image Processing in LEO Networks, *IEEE Transactions on Vehicular Technology*, 71(5):5661–5665 (K. Kim, J.-H. Lee, S. Jung, J. Kim, J.-H. Kim)
- [ISJ'22.03] LiteZKP: Lightning Zero-Knowledge Proof-based Blockchains for IoT and Edge Platforms, *IEEE Systems Journal*, 16(1):112–123 (E. Boo, J. Kim, J. Ko)
- [TVT'22.02] Quality-Aware Deep Reinforcement Learning for Streaming in Infrastructure-Assisted Connected Vehicles, *IEEE Transactions on Vehicular Technology*, 71(2):2002–2017 (W. Yun, D. Kwon, M. Choi, J. Kim, G. Caire, A. F. Molisch)
- [Nature'22.01] Feasibility Study of Multi-Site Split Learning for Privacy-Preserving Medical Systems under Data Imbalance Constraints in COVID-19, X-Ray, and Cholesterol Dataset, *Scientific Reports (Nature)*, 12:1534 (Y. J. Ha, G. Lee, M. Yoo, S. Jung, S. Yoo, J. Kim)
- ◀ 2021 ▶
- [RTIP'21.10] Adaptive and Stabilized Real-Time Super-Resolution Control for UAV-Assisted Smart Harbor Surveillance Platforms, *Journal of Real-Time Image Processing*, 18(5):1815–1825 (S. Jung, J. Kim)
- [ISJ'21.09] Intelligent Active Queue Management for Stabilized QoS Guarantees in 5G Mobile Networks, *IEEE Systems Journal*, 15(3):4293–4302 (S. Jung, J. Kim, J.-H. Kim)
- [Access'21.09] Spatio-Temporal Split Learning for Privacy-Preserving Medical Platforms: Case Studies with COVID-19 CT, X-Ray, and Cholesterol Data, *IEEE Access*, 9:121046–121059 (Y. J. Ha, M. Yoo, G. Lee, S. Jung, S. Choi, J. Kim, S. Yoo)
- [TVT'21.08] Infrastructure-Assisted On-Driving Experience Sharing for Millimeter-Wave Connected Vehicles, *IEEE Transactions on Vehicular Technology*, 70(8):7307–7321 (S. Jung, J. Kim, M. Levorato, C. Cordeiro, J.-H. Kim)
- [TMC'21.06] A Personalized Preference Learning Framework for Caching in Mobile Networks, *IEEE Transactions on Mobile Computing*, 20(6):2124–2139 (A. Malik, K. S. Kim, J. Kim, W.-Y. Shin)
- [TVT'21.06] Orchestrated Scheduling and Multi-Agent Deep Reinforcement Learning for Cloud-Assisted Multi-UAV Charging Systems, *IEEE Transactions on Vehicular Technology*, 70(6):5362–5377 (S. Jung, W. Yun, M. Shin, J. Kim, J.-H. Kim)
- [Access'21.06] Joint Mobile Charging and Coverage-Time Extension for Unmanned Aerial Vehicles, *IEEE Access*, 9:94053–94063 (S. Park, M. Choi, W.-Y. Shin, J. Kim)
- [ICTE'21.06] Truthful Electric Vehicle Charging via Neural-Architectural Myerson Auction, *ICT Express*, 7(2):196–199 (H. Lee, S. Jung, J. Kim)
- [PIEEE'21.05] Communication-Efficient and Distributed Learning Over Wireless Networks: Principles and Applications, *Proceedings of the IEEE*, 109(5):796–819 (J. Park, S. Samarakoon, A. Elgabri, J. Kim, M. Bennis, S.-L. Kim, M. Debbah)
- [TWC'21.04] Probabilistic Caching and Dynamic Delivery Policies for Categorized Contents and Consecutive User Demands, *IEEE Transactions on Wireless Communications*, 20(4):2685–2699 (M. Choi, A. F. Molisch, D.-J. Han, D. Kim, J. Kim, J. Moon)
- [JCN'21.04] Stabilized Adaptive Sampling Control for Reliable Real-Time Learning-based Surveillance Systems, *IEEE/KICS Journal of Communications and Networks*, 23(2):129–137 (D. Kim, S. Park, J. Kim, J. Y. Bang, S. Jung)
- [JCN'21.04] Dynamic Video Delivery using Deep Reinforcement Learning for Device-to-Device Underlaid Cache-Enabled Internet-of-Vehicle Networks, *IEEE/KICS Journal of Communications and Networks*, 23(2):117–128 (M. Choi, M. Shin, J. Kim)
- [JNCA'21.04] Contra-*: Mechanisms for Countering Spam Attacks on Blockchain's Memory Pools, *Journal of Network and Computer Applications*, 179:102971 (M. Saad, J. Kim, D. Nyang, D. Mohaisen)
- [ISJ'21.03] Multiscale LSTM-Based Deep Learning for Very-Short-Term Photovoltaic Power Generation Forecasting in Smart City Energy Management, *IEEE Systems Journal*, 15(1):346–354 (D. Kim, D. Kwon, L. Park, J. Kim, S. Cho)
- [ICTE'21.03] Distributed Deep Reinforcement Learning for Autonomous Aerial eVTOL Mobility in Drone Taxi Applications, *ICT Express*, 7(1):1–4 (W. Yun, S. Jung, J. Kim, J.-H. Kim)
- [BC'21.03] Empirically Comparing the Performance of Blockchain's Consensus Algorithms, *IET Blockchain*, 1(1):56–64 (A. Ahmad, A. Alabduljabbar, M. Saad, D. Nyang, J. Kim, D. Mohaisen)

◀ 2020 ▶

- [TWC'20.12] Joint Distributed Link Scheduling and Power Allocation for Content Delivery in Wireless Caching Networks, *IEEE Transactions on Wireless Communications*, 19(12):7810–7824 (M. Choi, A. F. Molisch, [J. Kim](#))
(IEEE ComSoc MMTC Best Journal Paper Award)
- [IOTJ'20.10] Multiagent DDPG-Based Deep Learning for Smart Ocean Federated Learning IoT Networks, *IEEE Internet of Things Journal*, 7(10):9895–9903 (D. Kwon, J. Jeon, S. Park, [J. Kim](#), S. Cho)
- [JCN'20.08] Self-Adaptive Power Control with Deep Reinforcement Learning for Millimeter-Wave Internet-of-Vehicles Video Caching, *IEEE/KICS Journal of Communications and Networks*, 22(4):326–337 (D. Kwon, [J. Kim](#), D. Mohaisen, W. Lee)
- [Access'20.06] Blind Signal Classification Analysis and Impact on User Pairing and Power Allocation in Nonorthogonal Multiple Access, *IEEE Access*, 8:100916–100929 (M. Choi [J. Kim](#))
- [TII'20.05] Cooperative Management for PV/ESS-Enabled Electric-Vehicle Charging Stations: A Multiagent Deep Reinforcement Learning Approach, *IEEE Transactions on Industrial Informatics*, 16(5):3493–3503 (M. Shin, D. Choi, [J. Kim](#))
- [ETRI'20.04] Simulation and Measurement: Feasibility Study of Tactile Internet Applications for mmWave Virtual Reality, *ETRI Journal*, 42(2):163–174 (W. Na, N.-N. Dao, [J. Kim](#), E.-S. Ryu, S. Cho)
- [ISJ'20.03] Towards Characterizing Blockchain-based Cryptocurrencies for Highly-Accurate Predictions, *IEEE Systems Journal*, 14(1):321–332 (M. Saad, J. Choi, D. Nyang, [J. Kim](#), A. Mohaisen) **(IEEE Systems Journal Best Paper Award)**
- [JCN'20.02] Numerical Approximation of Millimeter-Wave Frequency Sharing between Cellular Systems and Fixed Service Systems, *IEEE/KICS Journal of Communications and Networks*, 22(1):37–45 (S. Han, J.-W. Choi, [J. Kim](#))
- [JAIHC'20.01] A Novel Network Virtualization based on Data Analytics in Connected Environment, *Journal of Ambient Intelligence and Humanized Computing*, 11(1):75–86 (K.-H. N. Bui, S. Cho, J. Jung, [J. Kim](#), O.-J. Lee, W. Na)

◀ 2007–2019 ▶

- [TWC'19.12] Markov Decision Policies for Dynamic Video Delivery in Wireless Caching Networks, *IEEE Transactions on Wireless Communications*, 18(12):5705–5718 (M. Choi, A. No, M. Ji, [J. Kim](#))
- [TWC'19.10] Dynamic Power Allocation and User Scheduling for Power-Efficient and Delay-Constrained Multiple Access Networks, *IEEE Transactions on Wireless Communications*, 18(10):4846–4858 (M. Choi, [J. Kim](#), J. Moon)
- [IOTJ'19.10] Two-Stage IoT Device Scheduling with Dynamic Programming for Energy Internet Systems, *IEEE Internet of Things Journal*, 6(5):8782–8791 (L. Park, C. Lee, [J. Kim](#), A. Mohaisen, S. Cho)
- [TVT'19.10] Blind Signal Classification for Non-Orthogonal Multiple Access in Vehicular Networks, *IEEE Transactions on Vehicular Technology*, 68(10):9722–9734 (M. Choi, D. Yoon, [J. Kim](#))
- [TCAD'19.09] TEI-ULP: Exploiting Body Biasing to Improve the TEI-Aware Ultra-Low Power Methods, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 38(9):1758–1770 (W. Lee, T. Kang, J. Lee, K. Han, [J. Kim](#), M. Pedram)
- [WPC'19.08] Semantic Hashtag Relation Classification Using Co-occurrence Word Information, *Wireless Personal Communications*, 107(3):1355–1365 (S. Seo, J.-K. Kim, S.-I. Kim, J. Kim, [J. Kim](#))
- [TMC'19.07] Seamless Dynamic Adaptive Streaming in LTE/Wi-Fi Integrated Network under Smartphone Resource Constraints, *IEEE Transactions on Mobile Computing*, 18(7):1647–1660 (J. Koo, J. Yi, [J. Kim](#), M. A. Hoque, S. Choi)
- [TVT'19.05] Auction-Based Charging Scheduling With Deep Learning Framework for Multi-Drone Networks, *IEEE Transactions on Vehicular Technology*, 68(5):4235–4248 (M. Shin, [J. Kim](#), M. Levorato)
- [FGCS'19.04] Resource-Aware Relay Selection for Inter-Cell Interference Avoidance in 5G Heterogeneous Network for Internet of Things Systems, *Future Generation Computer Systems*, 93:877–887 (N.-N. Dao, M. Park, [J. Kim](#), J. Paek, S. Cho)
- [TETT'19.04] Thriving on Chaos: Proactive Detection of Command and Control Domains in Internet of Things-Scale Botnets using DRIFT, *Transactions on Emerging Telecommunications Technologies*, 30(4):e3505 (J. Spaulding, J. Park, [J. Kim](#), D. Nyang, A. Mohaisen)
- [CM'19.03] New Challenges of Wireless Power Transfer and Secured Billing for Internet of Electric Vehicles, *IEEE Communications Magazine*, 57(3):118–124 (L. Park, S. Jeong, D. S. Lakew, [J. Kim](#), S. Cho)
- [TIE'19.02] Joint Geometric Unsupervised Learning and Truthful Auction for Local Energy Market, *IEEE Transactions on Industrial Electronics*, 66(2):1499–1508 (L. Park, S. Jeong, [J. Kim](#), S. Cho)
- [IOTJ'18.12] Internet of Things for Smart Manufacturing System: Trust Issues in Resource Allocation, *IEEE Internet of Things Journal*, 5(6):4418–4427 (S. Jeong, W. Na, [J. Kim](#), S. Cho)
- [JSAC'18.11] SGCO: Stabilized Green Crosshaul Orchestration for Dense IoT Offloading Services, *IEEE Journal on Selected Areas in Communications*, 36(11):2538–2548 (N.-N. Dao, D.-N. Vu, W. Na, [J. Kim](#), S. Cho)
- [JSAC'18.06] Wireless Video Caching and Dynamic Streaming under Differentiated Quality Requirements, *IEEE Journal on Selected Areas in Communications*, 36(6):1245–1257 (M. Choi, [J. Kim](#), J. Moon)
- [Access'18.05] Soft Memory Box: A Virtual Shared Memory Framework for Fast Deep Neural Network Training in Distributed High Performance Computing, *IEEE Access*, 6:26493–26504 (S. Ahn, [J. Kim](#), E. Lim, S. Kang)
- [TVT'18.04] Adaptive Detector Selection for Queue-Stable Word Error Rate Minimization in Connected Vehicle Receiver Design, *IEEE Transactions on Vehicular Technology*, 67(4):3635–3639 (M. Choi, [J. Kim](#), J. Moon)
- [IOTJ'18.02] Energy-Efficient Mobile Charging for Wireless Power Transfer in Internet of Things Networks, *IEEE Internet of Things Journal*, 5(1):79–92 (W. Na, J. Park, C. Lee, K. Park, [J. Kim](#), S. Cho)
- [TII'17.12] Residential Demand Response for Renewable Energy Resources in Smart Grid Systems, *IEEE Transactions on Industrial Informatics*, 13(6):3165–3173 (L. Park, Y. Jang, S. Cho, [J. Kim](#))
- [IOTJ'17.10] Feasibility Study of 60 GHz Millimeter-Wave Technologies for Hyperconnected Fog Computing Applications, *IEEE Internet of Things Journal*, 4(5):1165–1173 ([J. Kim](#), W. Lee)
- [Access'17.09] A Software-based Monitoring Framework for Time-Space Partitioned Avionics Systems, *IEEE Access*, 5:19132–19143 (C. Shin, C. Lim, [J. Kim](#), H. Roh, W. Lee)
- [RTIP'17.09] QoS Optimal Real-Time Video Streaming in Distributed Wireless Image-Sensing Platforms, *Journal of Real-Time Image*

Processing, 13(3):547-556 (J. Kim, E. Ryu)

- [Access'17.08] Energy-Efficient Stabilized Automatic Control for Multicore Baseband in Millimeter-Wave Systems, *IEEE Access*, 5:16584–16591 (J. Kim, J.-J. Lee, J.-K. Kim, W. Lee)
- [Access'17.06] Adaptive Resource Balancing for Serviceability Maximization in Fog Radio Access Networks, *IEEE Access*, 5:14548–14559 (N.-N. Dao, J. Lee, D.-N. Vu, J. Paek, J. Kim, S. Cho, K. Chung, C. Keum)
- [IJAP'17.06] 60 GHz Modular Antenna Array Link Budget Estimation with WiGig Baseband and Millimeter-Wave Specific Attenuation, *International Journal of Antennas and Propagation*, 2017:9073465 (J. Kim, L. Xian, A. S. Sadri)
- [VTM'17.03] The Useful Impact of Carrier Aggregation: A Measurement Study in South Korea for Commercial LTE-Advanced Networks, *IEEE Vehicular Technology Magazine*, 12(1):55–62 (S. Lee, S. Hyeon, J. Kim, H. Roh, W. Lee)
- [TVT'16.12] Performance of Video Streaming in Infrastructure-to-Vehicle Telematic Platforms With 60-GHz Radiation and IEEE 802.11ad Baseband, *IEEE Transactions on Vehicular Technology*, 65(12):10111–10115 (J. Kim, S. Kwon, G. Choi)
- [Access'16.12] Numerical Simulation Study for Frequency Sharing between Micro-Cellular Systems and Fixed Service Systems in Millimeter-Wave Bands, *IEEE Access*, 4:9847–9859 (J. Kim, L. Xian, A. S. Sadri)
- [IJAP'16.12] Enhanced Next Generation Millimeter-Wave Multicarrier System with Generalized Frequency Division Multiplexing, *International Journal of Antennas and Propagation*, 2016:9269567 (H. Shimodaira, J. Kim, A. S. Sadri)
- [TON'16.08] Quality-Aware Streaming and Scheduling for Device-to-Device Video Delivery, *IEEE/ACM Transactions on Networking*, 24(4):2319–2331 (J. Kim, G. Caire, A. F. Molisch)
- [RTIP'16.08] Stochastic Stable Buffer Control for Quality-Adaptive HEVC Video Transmission in Enterprise WLAN Architectures, *Journal of Real-Time Image Processing*, 12(2):465–471 (J. Kim, E. Ryu)
- [TII'15.12] Energy-Efficient Dynamic Packet Downloading for Medical IoT Platforms, *IEEE Transactions on Industrial Informatics*, 11(6):1653–1659 (J. Kim)
- [TSMC'15.11] Stochastic Decision Making for Adaptive Crowdsourcing in Medical Big-Data Platforms, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 45(11):1471–1476 (J. Kim, W. Lee)
- [MTAP'15.10] Interference Impacts on 60 GHz Real-Time Online Video Streaming in Wireless Smart TV Platforms, *Multimedia Tools and Applications*, 74(19):8613–8629 (J. Kim, S.-N. Hong)
- [IJEC'15.07] Error Concealment Mode Signaling for Robust Mobile Video Transmission, *International Journal of Electronics and Communications*, 69(7):1070–1073 (E. Ryu, J. Kim)
- [TS'15.05] Dynamic Two-Stage Beam Training for Energy-Efficient Millimeter-Wave 5G Cellular Systems, *Telecommunication Systems*, 59(1):111–122 (J. Kim, S.-N. Hong)
- [CEE'15.04] Adaptive Buffer Control for Distributed Autonomous Robust Routing in Mobile Surveillance Robots, *Computers and Electrical Engineering*, 43:306–316 (J. Kim, S.-N. Hong)
- [JCN'14.10] Fast Millimeter-Wave Beam Training with Receive Beamforming, *IEEE/KICS Journal of Communications and Networks*, 16(5):512–522 (J. Kim, A. F. Molisch)
- [EL'14.10] Quality of Video Streaming in 38 GHz Millimetre-Wave Heterogeneous Cellular Networks, *IET Electronics Letters*, 50(21):1526–1528 (J. Kim, E. Ryu)
- [CL'14.09] Joint Coding and Stochastic Data Transmission for Uplink Cloud Radio Access Networks, *IEEE Communications Letters*, 18(9):1619–1622 (S.-N. Hong, J. Kim)
- [CL'14.07] A Low-Complexity Algorithm for Neighbor Discovery in Wireless Networks, *IEEE Communications Letters*, 18(7):1119–1122 (S.-N. Hong, J. Kim)
- [CL'14.03] Fast and Low-Power Link Setup for IEEE 802.15.3c Multi-Gigabit/s Wireless Sensor Networks, *IEEE Communications Letters*, 18(3):455–458 (J. Kim, A. Mohaisen, J.-K. Kim)
- [SCR'13.12] The Sybil Attacks and Defenses: A Survey, *Smart Computing Review*, 3(6):480–489 (A. Mohaisen, J. Kim)
- [TBC'13.09] Joint Scalable Coding and Routing for 60 GHz Real-Time Live HD Video Streaming Applications, *IEEE Transactions on Broadcasting*, 59(3):500–512 (J. Kim, Y. Tian, S. Mangold, A. F. Molisch)
- [EL'13.02] Distributed Stochastic Buffering for Enterprise WLAN Architectures, *IET Electronics Letters*, 49(4):302–304 (J. Kim, E. Ryu)
- [TCE'07.11] Movement-Aware Vertical Handoff of WLAN and Mobile WiMAX for Seamless Ubiquitous Access, *IEEE Transactions on Consumer Electronics*, 53(4):1268–1275 (W. Lee, E. Kim, J. Kim, I. Lee, C. Lee)
- [TCE'07.05] Coverage-Time Optimized Dynamic Clustering of Networked Sensors for Pervasive Home Networking, *IEEE Transactions on Consumer Electronics*, 53(2):433–441 (J. Kim, W. Lee, E. Kim, D.-W. Kim, H. Kim)
- [CL'07.01] Optimized Transmission Power Control of Interrogators for Collision Arbitration in UHF RFID Systems, *IEEE Communications Letters*, 11(1):22–24 (J. Kim, W. Lee, E. Kim, D. Kim, K. Suh)

■ IEEE ComSoc, VTS, and Computer Society Conference Contributions

- 50 papers at INFOCOM, ICDCS, GLOBECOM, ICC, WCNC, CCNC, PIMRC, VTC, APWCS, IEEE Quantum Week
- XX papers at other ComSoc, VTS, and Computer Society conferences

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IEEE Activities (Membership, Editorial Boards, and Services)

- Senior Member (2018–), Member (2006–2017)
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- Associate Editor (2025–), **IEEE Communications Surveys and Tutorials**
- Editor (2023–), **IEEE Internet of Things Journal**
- Associate Editor (2020–), **IEEE Transactions on Vehicular Technology**
- Guest Editor, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
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